SOLUTION

SECTION–I

1. Ans. (D)

$S_1P + S_2P = 2a$ & $S_1P - S_2P = 2A$

$S_1P = a + A$ & $S_2P = (a - A)$

quadratic equation is $x^2 - 2ax + (a^2 - A^2) = 0$

2. Ans. (C)

Let equation of line be $y = 2x + c$

solving it with $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

we get $b^2x^2 - a^2(2x + c)^2 = a^2b^2$

$\Rightarrow (b^2 - 4a^2)x^2 - 4a^2cx - a^2(b^2 + c^2) = 0$

For all 'c' this equation has atmost one solution so $b^2 = 4a^2$

Hence $e = \sqrt{1 + \frac{b^2}{a^2}} = \sqrt{5}$

3. Ans. (B)

Let the point be $(\alpha,\beta) \Rightarrow \beta = \alpha + 2$

Chord of contact of hyperbola is $T = 0$

$$\Rightarrow \frac{\alpha x}{6} - \frac{\beta y}{2} = 1 \Rightarrow \frac{\alpha x}{6} - \frac{(\alpha+2)y}{2} = 1$$

$$\Rightarrow \alpha\left(\frac{x}{6} - \frac{y}{2}\right) - \left(\frac{2y}{2} + 1\right) = 0$$

Since this passes through (λ,μ)

$$\therefore \frac{\lambda}{6} - \frac{\mu}{2} = 0 \text{ and } \frac{2\mu}{2} + 1 = 0 \Rightarrow \frac{\lambda}{\mu} = 3$$

4. Ans. (D)

The chord of contact of (2, 3) is $3y = 2(x + 2)$

solving it with $y^2 = 4x$

$y^2 = 2(3y - 4) \Rightarrow y^2 - 6y + 8 = 0 \Rightarrow y = 2, 4$

so $x = 1$ & 4 respectively so points are (1, 2) and (4, 4).

5. Ans. (A)

Equation of the normal to the parabola $y^2 = 12x$ is given by

$y = mx - 6m - 3m^3$

It passes through (1, 2) so

$2 = m - 6m - 3m^3$

$\Rightarrow 3m^3 + 5m + 2 = 0$

$m_1m_2 + m_2m_3 + m_3m_1 = 5/3$ ...(i)

$m_1m_2m_3 = -2/3$ ...(ii)

(i) divided by (ii)

$$\frac{1}{m_1} + \frac{1}{m_2} + \frac{1}{m_3} = \frac{-5}{2}.$$

6. Ans. (C)

Any tangent to $y = x^2$ can be written as

$y = mx - \frac{1}{4}m^2$ if it touches 2nd, then

$$mx - \frac{m^2}{4} = x^2 - 2x + 2 \Rightarrow$$

$$x^2 - (2 + m)x + \left(2 + \frac{m^2}{4}\right) = 0$$

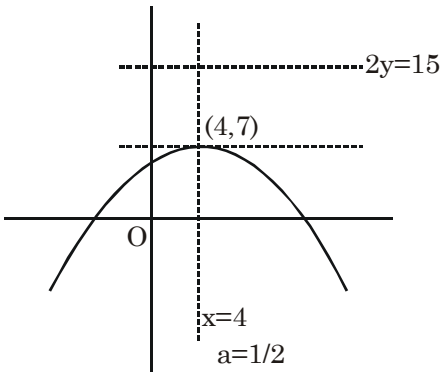
$D = 0 \Rightarrow (m+2)^2 - 4\left(2 + \frac{m^2}{4}\right)$

$4m - 4 = 0 \Rightarrow m = 1 \Rightarrow 4x - 4y - 1 = 0$

7. Ans. (A)

$x^2 - 2.4x + 16 = 16 - 2 - 2y$

$(x - 4)^2 = -2(y - 7)$



8. Ans. (B)

Focus $\equiv (2, 0)$, point $\equiv (2t^2, 4t)$

$$\Rightarrow h = \frac{2 + 2t^2}{2}$$

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$$k = \frac{4t+0}{2}$$

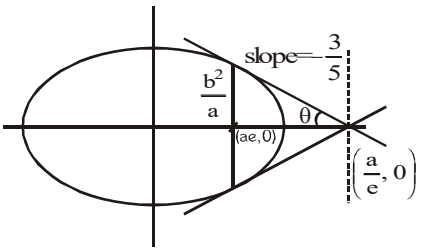
$\Rightarrow$ $h = 1 + t^2$ & $k = 2t$

$\Rightarrow$ $h = 1 + \frac{k^2}{4}$ $\Rightarrow$ $y^2 = 4x - 4$

9. Ans. (D)

Point of intersection of tangents is $\left(\frac{20}{3}, 0\right)$ which lie on the corresponding directrix of the ellipse

$\therefore$ $\frac{a}{e} = \frac{20}{3}$



$$\tan\theta = \frac{b^2}{a\left(\frac{a}{e} - ae\right)} = \frac{b^2 e}{a^2(1-e^2)} = e$$

$\frac{3}{5} = e$ $\therefore$ $a = \frac{20}{3} \times \frac{3}{5} = 4$

$$b^2 = 16\left(1 - \frac{9}{25}\right) = \frac{16 \times 16}{25}$$

$$\ell_{LR} = \frac{2b^2}{a} = \frac{2 \times 256}{4 \times 25} = 5.12$$

10. Ans. (D)

The vertices of the square will lie on director's circle of ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$

$\Rightarrow$ vertices of square lies on $x^2 + y^2 = 16 + b^2$

$\therefore$ $2(\text{side of square})^2 = (\text{diameter of director's circle})^2$

$\Rightarrow$ Radius of director's circle = 5 $\Rightarrow$ $16 + b^2 = 25$ or $b = \pm 3$

$\therefore$ Area of ellipse = $\pi(4)(3)$ sq. units

$= 12\pi$ sq. units

11. Ans. (B,D)

Let the common tangent be

$y = mx \pm \sqrt{12m^2 - 12}$ solving it with

$xy = 8$ $\Rightarrow$ $mx^2 \pm \sqrt{12(m^2-1)}x - 8 = 0$

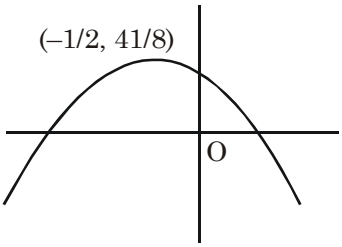
$D = 0$ $\Rightarrow$ $12(m^2 - 1) + 32m = 0$

$3m^2 + 8m - 3 = 0$ $\Rightarrow$ $m = -3, \frac{1}{3}$

$y = -3x \pm 4\sqrt{6}$ $m = \frac{1}{3}$ rejected

12. Ans. (A,D)

$(f(0) - 5)^2 + (f(1) - 4)^2 + (f(2) - 2)^2 = 0$

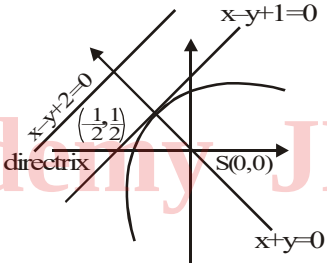


$f(0) = 5,\ f(1) = 4,\ f(2) = 2$

$f(x) = -\frac{x^2}{2} - \frac{x}{2} + 5$

13. Ans. (A,C)

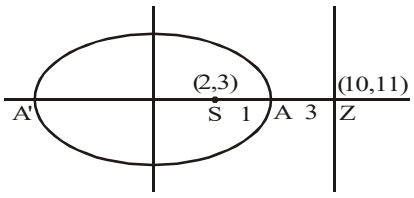
$(x + y)^2 = 4(x - y + 1)$



$$2\left(\frac{x+y}{\sqrt{2}}\right)^2 = 4\sqrt{2}\left(\frac{x-y+1}{\sqrt{2}}\right)$$

$$\left(\frac{x+y}{\sqrt{2}}\right)^2 = 2\sqrt{2}\left(\frac{x-y+1}{\sqrt{2}}\right)$$

14. Ans. (A,B,C)



$A \equiv \left(\frac{10+6}{4}, \frac{11+9}{4}\right) = (4,5)$

$A' \equiv \left(\frac{10-6}{1-3}, \frac{11-9}{1-3}\right) = (-2,-1)$

Major axis, $AA' = \sqrt{36+36} = 6\sqrt{2}$

Minor axis $2b = 2a\sqrt{1-e^2} = 2\left(3\sqrt{2}\right)\sqrt{1-\frac{1}{9}} = 8$

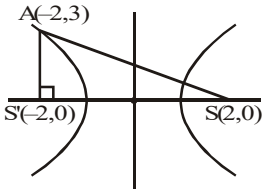
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Latus rectum = $\frac{2b^2}{a} = \frac{2\times16}{3\sqrt{2}} = \frac{16\sqrt{2}}{3}$

15. Ans. (A,B,C,D)

$S \equiv (2, 0)$, $S' \equiv (-2, 0)$



Using reflection property of hyperbola,

S'A is incident ray.

Equation of incident ray

S'A is $x = -2$

Equation of reflected ray

SP is $3x + 4y = 6$.

Now $2ae = 4 \Rightarrow ae = 2$ ....(i)

Point $(-2, 3)$ lies on hyperbola,

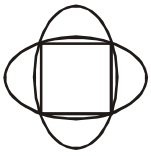
$\therefore \frac{4}{a^2} - \frac{9}{b^2} = 1 \Rightarrow \frac{4}{a^2} - \frac{9}{4-a^2} = 1$

on solving it we get $a = 4$ (reject), $a = 1$ .....(ii)

$\therefore$ Using (i) & (ii), we get $e = 2$

length of latus rectum $= 2a(e^2 - 1) = 6$

16. Ans. (A,C)



The given points lie on

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ & $\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$

The given points lie on a square & no parabola can pass through all vertices of a square

Paragraph for Question 17 & 18

17. Ans. (B)

Minimum value occurs at $x = 2$

so $f(2) = |a - 2| = 1 \Rightarrow a = 3, 1$

$\Rightarrow a = 3$ (as > 2)

$\Rightarrow$ Ellipse is $\frac{x^2}{9} + \frac{y^2}{4} = 1$

& $e = \sqrt{1 - \frac{4}{9}} = \frac{\sqrt{5}}{3}$

18. Ans. (D)

Let tangent AB is $\frac{x\cos\theta}{3} + \frac{y\sin\theta}{2} = 1$

AB passes through P(1, 2)

$\Rightarrow \frac{\cos\theta}{3} + \sin\theta = 1$

$\theta = \frac{\pi}{2}$ is one solution

also $\frac{1}{3} + \tan\theta = \sec\theta \Rightarrow \sec\theta - \tan\theta = \frac{1}{3}$ ....(i)

& $\sec\theta + \tan\theta = 3$ .....(ii)

$\Rightarrow \sec\theta = \frac{5}{3}$ & $\tan\theta = \frac{4}{3}$

Hence none of the options necessarily are true, so (D)

Paragraph for Question 19 & 20

19. Ans. (D)

Focus of C_2 is $(\sqrt{2}, \sqrt{2})$ &

asymptote of C_1 is $y = -x$ & $y = x$.

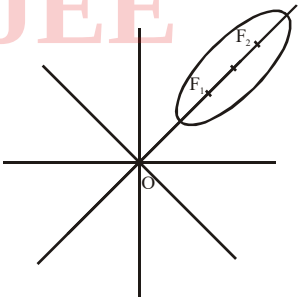
$\therefore$ Focus of C_3 is $F_1(\sqrt{2}, \sqrt{2})$ directrix is $y = -x$

for curve C_3

$a\left(\frac{1}{e} - e\right) = 2$

$\frac{4}{3}\left(\frac{1}{e} - e\right) = 2$

$\Rightarrow e = \frac{1}{2}$

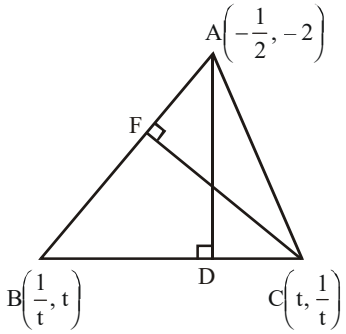


20. Ans. (A)

$A\left(-\frac{1}{2}, -2\right)$

Let $B\left(\frac{1}{t}, t\right)$,

then $C\left(t, \frac{1}{t}\right)$



Slope of BC $= -1 \Rightarrow$ Slope of AD $= 1$

Equation of AD : $y + 2 = x + \frac{1}{2}$

or $2x - 2y = 3$ .....(1)

Now slope of AB $= 2t$

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$\Rightarrow$ Slope of $CF = -\frac{1}{2t}$

$\Rightarrow$ equation of CF : $y - \frac{1}{t} = -\frac{1}{2t}(x - t)$

or $x + 2ty = t + 2$ .....(2)

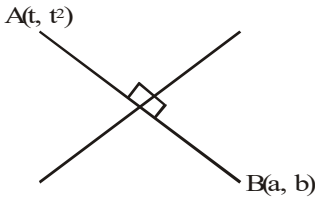
Solving (1) & (2), we get

$\therefore$ orthocentre is $\left(2, \frac{1}{2}\right)$

SECTION–III

1. Ans. 2

General point on $y = x^2$ in (t, t^2) let us find the image of (t, t^2) in $x + y = 1$



$$\frac{a+t}{2} + \frac{b+t^2}{2} = 1$$

$a + t + b + t^2 = 2$ ...(1)

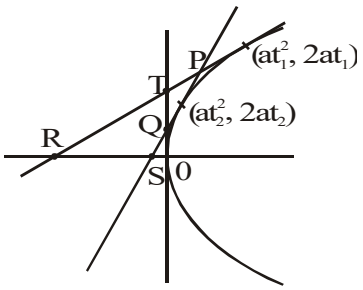
and $\frac{b-t^2}{a-t} = 1$ ...(2)

eliminate t from (1) & (2)

$a + b^2 = 2b$

$\Rightarrow$ $x + y^2 = 2y$ $\Rightarrow$ $k = 2$

2. Ans. 1



The equation of first tangent is $t_1y = x + at_1^2$

$T(0, at_1)$ & $R(-at_1^2, 0)$

For second tangent

$Q(0, at_2)$ & $S(-at_2^2, 0)$

For concyclic points (OT) (OQ) = (OR) (OS)

$\Rightarrow t_1t_2 = 1$

Point $x = at_1t_2$

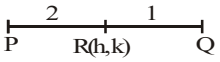
$\Rightarrow$ $x = a$

$\Rightarrow$ $\lambda_1 = 1$

3. Ans. 9

Equation of tangent is $\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1$

$P\left(\frac{a}{\cos\theta}, 0\right)$ & $Q\left(0, \frac{b}{\sin\theta}\right)$



$h = \frac{a}{3\cos\theta}$ & $k = \frac{2b}{3\sin\theta}$

$\cos\theta = \frac{a}{3h}$ & $\sin\theta = \frac{2b}{3k}$

$\frac{a^2}{h^2} + \frac{4b^2}{k^2} = 9$ $\Rightarrow$ $\frac{a^2}{x^2} + \frac{4b^2}{y^2} = 9$

4. Ans. 1

A tangent to $\frac{x^2}{16} - \frac{y^2}{9} = 1$ is

$$y = m_1x \pm \sqrt{16m_1^2 - 9} \quad .....(i)$$

The other hyperbola is $\frac{x^2}{-9} - \frac{y^2}{-16} = 1$

then tangent to it is

$y = m_2x \pm \sqrt{-9m_2^2 - (-16)}$ .....(ii)

(i) & (ii) are same.

$\Rightarrow$ $m_1 = m_2$ & $16m_1^2 - 9 = -9m_2^2 + 16$

$\Rightarrow$ $25m_1^2 = 25$ $\Rightarrow$

$m_1^2 = 1$ $\Rightarrow$ $m_1 = \pm 1$

$\therefore$ $m = 1$

5. Ans. 5

Assume (h,k) be the middle point of chord

$\therefore$ Its equation is

$hx - ky = h^2 - k^2 \Rightarrow y = \frac{h}{k}x + \frac{k^2 - h^2}{k}$

Now using $c = \frac{a}{m}$, we get

$\frac{k^2 - h^2}{k} = \frac{5}{h}.k$

$\Rightarrow$ $x^3 = y^2(x - 5)$

$\therefore$ $a = 5$

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